Solid State Physics — Week 11 Assignment Questions & Answers

Question 101

Diamagnetism is exhibited by

(a) Only those materials which do not possess permanent magnetic moments.

(b) All materials

(c) Only those materials which possess permanent magnetic moments

(d) Only those materials which have strong internal fields

Accepted answer: (b)

Question 102

Paramagnetic materials when placed in a magnetic field get

(a) Feebly attracted

(b) Feebly repelled

(c) Strongly repelled

(d) Strongly attracted

Accepted answer: (a)

Question 103

The net spin magnetic moment of Fe^{3+} ion (inner core $3d^5$) is

(a) $5\mu_B$

(b) $4\mu_B$

(c) $3\mu_B$

(d) $2\mu_B$

Accepted answer: (a)

Question 104

Susceptibility of diamagnetic material

(a) Varies directly with temperature (T)

(b) Varies as $\frac{1}{T-\theta}$ where θ is the Curie temperature

(c) Varies as $\frac{1}{T}$

(d) Is independent of temperature

Accepted answer: (d)

Question 105

Calculate the diamagnetic susceptibility (susceptibility per kg. mol or molar susceptibility) of He atom given its average atomic radius is 0.53 Å.

(a) $1.9937 \times 10^{-8}/mol$

(b) $2.1322 \times 10^{-10}/mol$

(c) $4.2144 \times 10^{-12}/mol$

(d) $5.2723 \times 10^{-14}/mol$

Accepted answer: (a)

Question 106

Magnetic susceptibility of aluminum at room temperature is 0.82×10^{-8}. Calculate its magnetization when a magnetic field of 0.25 Tesla is applied.

(a) $2.14 \times 10^{-6} A/m$

(b) $3.37 \times 10^{-4} A/m$

(c) $1.63 \times 10^{-3} A/m$

(d) $6.12 \times 10^{-2} A/m$

Accepted answer: (c)

Question 107

Calculate the saturation magnetization of Gadolinium given that Bohr magneton $\mu_B = 9.27 \times 10^{-24} A/m^2$, atomic number = 64, atomic weight = 157.25, density = $7.86 \times 10^3\ kg/m^3$ and electronic configuration for inner core $(4f^7 5s^2 5p^6 5d^1)$.

(a) $1.13 \times 10^4 A/m$

(b) $2.23 \times 10^6 A/m$

(c) $4.13 \times 10^8 A/m$

(d) $5.27 \times 10^{10} A/m$

Accepted answer: (b)

Question 108

A paramagnetic salt contains $10^{26}\ ions/m^3$ with magnetic moment $0.1\mu_B$. Calculate the magnetization (M) of the salt in a field of 1T at 300K using the following relation $\left[M = N\mu \tanh\left(\frac{\mu B}{K_B T}\right)\right]$. Where N is the number of atoms per unit volume, μ_B is the Bohr magneton and K_B is the Boltzmann constant.

(a) $0.02\ A/m$

(b) $0.03\ A/m$

(c) $0.04\ A/m$

(d) $0.05\ A/m$

Accepted answer: (a)

Question 109

A system of electron spins is placed in a magnetic field of $2\ Wb/m^2$ at a temperature T. Under equilibrium condition the number of spins parallel to the magnetic field is twice as large as the number of antiparallel spins. Determine the temperature T. Bohr magneton $\mu_B = 9.27 \times 10^{-24} A/m^2$ and Boltzmann constant $K_B = 1.38 \times 10^{-23} J/K$.

(a) $3.87\ K$

(b) $4.24\ K$

(c) $5.27\ K$

(d) $9.17\ K$

Accepted answer: (a)

Question 110

The magnetic susceptibility of iron measured at 900°C is 2.5×10^{-4}. Find the Curie temperature of iron. It is given that the Curie constant $C = 3.25 \times 10^{-2}/K$.

(a) $1043\ K$

(b) $1423\ K$

(c) $1753\ K$

(d) $1954\ K$

Accepted answer: (a)

Question 111

Nickel is FCC and its lattice parameter is 3.52 Å. The magnetic moment of Nickel atom is $0.6\mu_B$. Calculate the saturation magnetization of Nickel.($\mu_B = 9.27 \times 10^{-24} A/m^2$)

(a) 5.1×10^5 A/m

(b) 4.1×10^4 A/m

(c) 3.1×10^3 A/m

(d) 2.8×10^2 A/m

Accepted answer: (a)